NAME: ____________________ **PERIOD:** ________ **DATE:** ____________

A Percent Needs Its Base

AP Statistics · Unit 1 · Topic 1.3 · B: 2026-09-11 · E: 2026-09-14

[STAR] TODAY'S PROBLEM

Two clubs asked the student council for money to run a Saturday session. Each club surveyed *all* of its members: "Would you attend a Saturday session?" The table shows every response as a count.

At the council meeting, the **Robotics club** presented a **frequency table** (counts) and the **Chess club** presented a **relative-frequency table** (percents). Each club chose the table that made its case look best.

"Would you attend a Saturday session?" (every member surveyed; counts)

Club	**Yes**	**No**	**Total**
Chess	15	3	18
Robotics	27	18	45

FIRST TAKE — before the video (5 min)

Which club has more interest in a Saturday session — Chess or Robotics? One sentence and one number.

[BOOK] TABLES FOR A CATEGORICAL VARIABLE (keep on the page)

A **frequency table** gives the **count** in each category. A **relative-frequency table** gives the **proportion** (or percent) in each category: relative frequency $= \frac{\text{count in the category}}{\text{total}}$, times 100 for a percent. The percents in one table add to 100%.

A percent needs its base. "83% said yes" means nothing until you know 83% *of how many*. Counts answer "how many?"; percents answer "what share?" — use the table that answers the question asked.

Finish after the video:

DOK 1 **(a)** How many Robotics members said Yes? Which club has more members, and how many more?

DOK 2 **(b)** Build the relative-frequency table: for each club, the percent of its members who said Yes and the percent who said No (nearest whole percent). Which club has the higher percentage saying Yes?

DOK 3 **(c)** The council asks: "Which club has **more student interest** in a Saturday session?" Decide which table — counts or percents — answers *that* question, and cite the specific numbers that settle it. Then explain what each club's chosen table **hides** from a reader: what does the Robotics club's frequency table hide, and what does the Chess club's percent table hide?

Frame: "More student interest" is answered by the ____________ table because the question asks ______________________________.

Frame: The Chess club's percents hide that ___________________________; the Robotics club's counts hide that ___________________________.

EXIT — turn this in

One thing the video changed about my first take: ____